ABSTRACT

In today's fast-paced world, music serves as a powerful medium to influence emotions and enhance moods. This project introduces an **Intelligent Mood-Based Music Player**, which leverages **facial recognition technology** to detect a user's emotional state and automatically curate a personalized playlist. By analyzing **facial expressions**, the system categorizes emotions such as **happy, sad, angry, or neutral**. Based on the detected mood, it selects and plays songs that either complement or uplift the user's emotional state, providing a seamless and immersive listening experience.

This approach enhances user engagement by eliminating the need for manual song selection and delivering music that resonates with their feelings. The system personalizes music recommendations and creates a more **intuitive and user-friendly** interaction. By integrating **emotion recognition and music streaming**, this project transforms how users experience and interact with music, making it a **unique and engaging** entertainment solution.

SYSTEM ANALYIS

EXISTING SYSTEM

In traditional music players, users manually **search for, select, and play songs** based on their mood or preference. Many popular music streaming platforms offer **mood-based playlists**, but they require users to manually navigate through categories and choose songs. Some advanced music applications recommend songs based on **listening history** rather than real-time emotional state.

Additionally, some existing systems use **manual mood selection**, where users must indicate their emotions before receiving recommendations. However, this approach lacks automation and **does not dynamically adjust** to real-time emotional changes. Furthermore, most applications **do not integrate facial expression recognition**, making mood-based music selection an inconvenient and time-consuming process.

PROPOSED SYSTEM

The Mood-Based Music Player introduces an automated, intelligent approach to music selection using real-time facial expression detection. By leveraging CameraX for facial capture and emotion classification, the system eliminates the need for manual song selection. The application detects the user's mood and instantly plays a curated playlist that aligns with their emotional state.

Key Enhancements Over Existing Systems:

Automated Mood Detection – Uses the device's front camera to analyze facial expressions in real-time.

Dynamic Playlist Selection – Adjusts song playback automatically based on detected mood (Happy, Sad, or Neutral).

Personalized Experience – Allows users to add, remove, and manage their mood-based playlists.

Offline Music Support – Enables playback of locally stored songs without an internet connection.

Smooth User Experience – Provides an intuitive UI with seamless transitions and animations.

This system enhances user convenience, reduces decision fatigue, and ensures an engaging, personalized music experience tailored to the listener's emotional state.

FEASIBILITY STUDY

Technical Feasibility

Technology Stack

- Programming Language: Java (Android SDK)
- UI Framework: XML for UI layouts
- Media Handling: Android MediaPlayer
- Storage: SharedPreferences for playlist persistence
- Mood Detection: Camera integration using CameraX
- Device Compatibility: Android 8.0 (API Level 26) and above

System Requirements

- Hardware: Smartphone with a front-facing camera
- Software: Android OS, supporting CameraX and MediaPlayer
- Permissions: Camera and Storage access

Challenges & Risks

- Accuracy of Mood Detection: Relies on facial expressions, which can be subjective.
- Battery & Performance: Continuous media playback and animation might drain battery quickly.
- Storage Management: Handling large playlists without overloading SharedPreferences.

Economic Feasibility

Development Cost

- Resources Needed: Developer(s) with Android expertise
- Development Tools: Android Studio (free), testing devices
- Third-Party Services: Possible use of APIs for advanced mood recognition
- Marketing & Distribution: Play Store listing, potential in-app purchases or ads

Revenue Model

- Freemium Model: Basic features free, premium version with added functionalities
- In-App Ads: Monetization through advertisements
- Subscription Model: Unlimited access to premium playlists and customization

Operational Feasibility

User Perspective

- Ease of Use: Simple UI with intuitive buttons for song control
- Customization: Users can manually add or remove songs
- Reliability: Ensuring smooth playback and minimal crashes

Maintenance & Support

- Regular Updates: Bug fixes and new features
- User Feedback Mechanism: Integration of feedback options for improvements

Legal Feasibility

- Data Privacy: Handling user data (mood detection) responsibly
- Permissions Compliance: Requesting and explaining storage & camera access
- Copyright Issues: Users should have legal access to stored songs

SYSTEM REQUIREMENTS SPECIFICATION

HARDWARE SYSTEM

Camera: A high-resolution camera or smartphone camera to capture facial expressions in real-time.
Processor: High-performance CPU/GPU to process video frames for face detection and emotion recognition in real-time.
Speakers/Headphones: Audio output devices for playing music according to the detected mood.

SOFTWARE SYSTEM

Camera: A high-resolution camera or smartphone camera to capture facial expressions in real-time.
Processor: High-performance CPU/GPU to process video frames for face detection and emotion recognition in real-time.
Speakers/Headphones: Audio output devices for playing music according to the detected mood.

ABOUT SOFTWARE

Android Studio

Android Studio is the official Integrated Development Environment (IDE) for Android app development, developed by Google. It provides powerful tools for designing, coding, debugging, and testing applications. Key features of Android Studio include

- **Code Editor:** Supports Java, Kotlin, and XML with intelligent code completion.
- **Layout Editor:** A drag-and-drop UI designer for building Android user interfaces.
- **Android Emulator:** Allows developers to test applications on virtual devices.
- **Gradle Build System:** Automates the build process and manages dependencies.
- **Debugging Tools:** Helps developers analyze app performance and identify issues.

Android SDK (Software Development Kit)

The **Android SDK** is a set of tools and libraries required to develop Android applications. It includes

- **Android API Libraries:** Provides pre-built functions for accessing device features like camera, storage, and sensors.
- **Android Debug Bridge (ADB):** A command-line tool used for debugging and interacting with connected Android devices.
- **SDK Tools:** Includes tools like **AVD Manager** (for creating virtual devices) and **SDK Manager** (for downloading required packages).
- **Platform APIs:** Enables developers to create apps that are compatible with different Android versions.

CameraX

CameraX is an Android Jetpack library that simplifies camera app development. It provides an easy-to-use API for integrating camera functionalities with minimal code. Features include:

- **Live Preview:** Displays real-time camera feed.
- **Image Capture:** Captures high-quality images.
- **Video Recording:** Supports video capture with advanced processing.
- **Face Detection & Image Analysis:** Allows integration with ML Kit for real-time face and emotion detection.
- **Device Compatibility:** Works across a wide range of Android devices with consistent performance.

MediaPlayer

The **MediaPlayer** class in Android is used to play audio and video files. It supports various file formats, including MP3, AAC, and MP4. Key features include

· **Streaming Support:** Can play media from the internet using URLs.

· **Playback Controls:** Start, pause, stop, and seek functionalities.

· **Background Playback:** Allows audio to continue playing even when the app is minimized.

· **Integration with UI:** Works with buttons, sliders, and notifications for a seamless media experience.

SharedPreferences

SharedPreferences is a lightweight data storage solution used for storing key-value pairs. It is commonly used for

- **Saving User Preferences:** Stores settings like dark mode, language selection, and sound settings.
- **Session Management:** Saves login credentials or temporary data.
- **App State Preservation:** Retains user data even after the app is closed.
- **Quick Data Retrieval:** Provides fast access to stored values without a database.

Example

```
SharedPreferences preferences = getSharedPreferences("MyPrefs", MODE_PRIVATE);

SharedPreferences.Editor editor = preferences.edit();

editor.putString("username", "JohnDoe");

editor.apply();
```

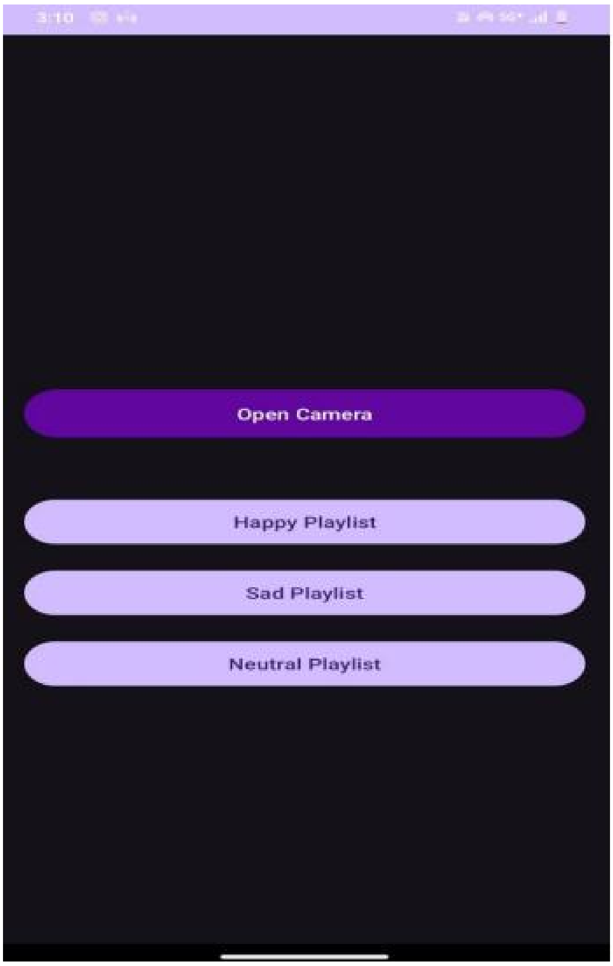
Syntax

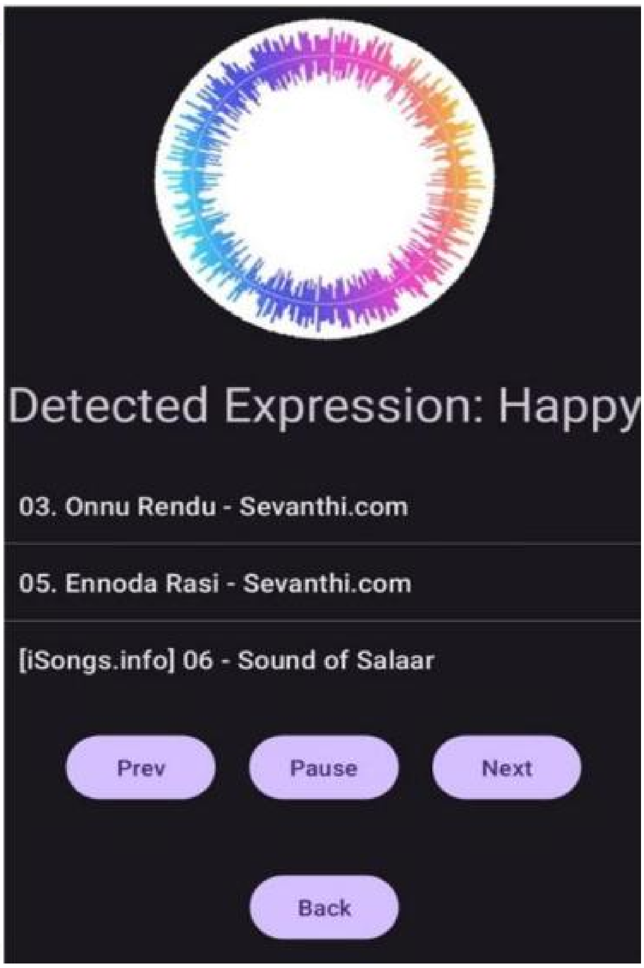
```
String username = preferences.getString("username", "DefaultUser");
```

XML (Extensible Markup Language)

XML is used in Android for designing UI layouts and defining configurations. It plays a key role in:

- **Layout Design:** Used in **activity_main.xml** to create UI elements.
- **Drawable Resources:** Defines shapes, colors, and styles for UI components.
- **Manifest File (AndroidManifest.xml):** Specifies app permissions, components, and metadata.
- **Navigation and Animation:** Used in **nav_graph.xml** for app navigation and **anim.xml** for animations.


Open Camera
Happy Playlist
Sad Playlist
Neutral Playlist


Detected Expression: Happy
03. Onnu Rendu - Sevanthi.com
05. Ennoda Rasi - Sevanthi.com
[iSongs.info] 06 - Sound of Salaar
Prev
Pause
Next
Back